

TimeLightHub — Technical addendum to the Phase-1 proposal: readout wavelength and read-disturb

Samer Beyrouti · Connect@TimeLightHub.com · 4 October 2026 · Applies to the Phase-1 Protocol (v1.0, September 2026) and the Brief Proposal (22 September 2026). It supersedes the read step in both.

The correction. The earlier documents specified a 405 nm read and called it non-destructive. That was an error on my side. In a diarylethene, any probe light absorbed by either isomer carries a finite quantum yield for the photoreaction, so an absorption read disturbs the stored state, and at 405 nm many open-ring forms still absorb and ring-close. Reading through the refractive-index change at a wavelength away from the absorption bands is the established answer: Tanio and Irie (1994) and Ebisawa et al. (1994) in polymer waveguides and a Mach-Zehnder, and Bilodeau et al. (Optica, 2024) on a silicon-nitride ring. The read wavelength is therefore a **measured design variable**, not a fixed number, and read-disturb must be measured explicitly.

1. Revised read strategy: three regimes, measured side by side

Regime	Probe	Role
R1 · Off-resonance phase read	A wavelength at which neither isomer absorbs appreciably, chosen from the laboratory's measured spectra of its own DAE: roughly 700–850 nm, or the telecom bands at 1310 or 1550 nm, which give the lowest disturb and use standard parts. The stored state is detected through the refractive-index difference between isomers, as a phase shift.	The architecture's intended read mode. The probe is absorbed only through the residual tail of the bands, which M3 quantifies.
R2 · Low-disturb absorption read	The long-wavelength tail of the closed-ring band (for example 630–680 nm), where contrast is high and the cycloreversion yield is low.	Reference with a large signal and a small, measurable disturb.
R0 · 405 nm read	As originally specified.	Worst-case reference only.

Write (about 375 nm, or the laboratory's nearest UV line) and reset (532 nm) are unchanged. Both can be delivered from above, as flood or focused illumination, so that UV does not need to be guided.

2. Revised measurements

Quantity	How it is obtained
M1 · Readout latency	Shortest probe pulse that resolves the two states at a stated signal-to-noise ratio, for R1 and for R2.
M2 · Energy per bit	Write and reset: minimum dose for a full, repeatable switch × cell area. Read: probe energy per resolved read at the same signal-to-noise ratio, for R1 and R2, stated with its boundary: at the detector, launched, and wall-plug.
M3 · Read-disturb (new)	Loss of stored state per unit of probe dose, for R0, R1 and R2: the accumulated probe energy through the cell at which the contrast (R2) or the phase difference (R1) falls by 10%, converted to an equivalent number of reads. A dose gate is used because a weight read at gigahertz rates sees 10 ⁶ reads in a millisecond.
Supporting	Refractive-index difference between isomers versus wavelength, Δn(λ); absorption of both isomers, α(λ); cycle endurance; dark retention; waveguide loss in both states. Controls: an undoped host film under the same exposure (host photo-index change) and a thermal-drift reference.

Revised success gate. Go: repeatable write, read and reset; in R1 less than 10% change of the stored state after an accumulated probe dose equivalent to at least 10¹² reads; read energy inside the architecture's window, which is defined per weight read (one stored value, with the number of levels per cell stated), not per bit. Learn: only R2 is usable, and its reads-per-refresh figure is reported, which turns the memory into a refreshed one and revises the architecture's energy claim with real numbers. Stop: no usable index contrast off resonance in the chosen host.

What read energy to expect — a first-order estimate, to be replaced by the measurement. In R1 the probe is absorbed only through the residual tail, so the read energy is set mainly by detection. The quantum floor for a binary read is $\delta\phi \approx 1/\sqrt{N}$: with $\Delta n \approx 10^{-3}$ (reported values for DAE-doped polymers span roughly 10⁻⁴ to 10⁻²), $L = 100\text{ }\mu\text{m}$, a modal overlap $\Gamma \approx 0.2$ and $\lambda = 780\text{ nm}$, $\Delta\phi = 2\pi\Gamma\Delta nL/\lambda \approx 0.16\text{ rad}$, and $N \approx 4,000$ detected photons, about 1 fJ, at a signal-to-noise ratio of 10. **That is a floor, not a forecast.** A practical gigahertz receiver is thermal-noise limited and needs roughly 10–50 fJ at the detector; with optical losses and laser efficiency, the electrical cost of the light is expected between a few tenths of a picojoule and a few picojoules per individually detected bit. We therefore expect Phase 1 to report about **0.03–0.3 pJ** of probe energy per resolved read on a bench receiver. Whether the architecture's window is met depends on Δn at the read wavelength, on the receiver, and on how many weights share one detection in an analog sum; the measurement reports all three.

3. What changes for the laboratory

- **One additional probe source** in the red or near infrared (633, 780 or 850 nm), or at 1310 or 1550 nm, in place of the 405 nm read.
- **An interferometric readout.** On a polymer-waveguide platform a pair of Y-branches forms a Mach-Zehnder with the DAE-doped section in one arm; a ring resonator or a simple Fabry-Pérot fringe shift serves equally. On planar films, ellipsometry or a prism coupler gives Δn(λ) directly.
- **A read-disturb run:** a continuous or pulsed probe of known power, with the contrast measured at intervals of accumulated dose. No new equipment.
- Time-resolved detection, the write and reset sources, the films and the waveguides are as in the proposal.

4. Two questions on which your view would help

- From the spectra of your own diarylethene in SU-8, which probe wavelength would you choose for R1 and for R2?
- Has the index difference between isomers of your films been measured, and at which wavelengths?